

ToxPipe: Leveraging Expert Entrained AI and Cloud Computing for Advanced Computational Toxicology

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Abstract

The Division of Translational Toxicology at NIEHS introduces ToxPipe, a pioneering project aiming to harness expert entrained AI-based systems and cloud computing to revolutionize the analysis and interpretation of toxicological properties of compounds. ToxPipe repurposes advanced AI models like JARVIS and Auto-GPT to facilitate the exploration of toxicologically-relevant data through natural language instructions, enabling a semi-autonomous approach to toxicological characterization and data integration. This project is grounded in the success of expert entrained AI models like Auto-GPT and JARVIS, which employ OpenAI's GPT-based models as controllers to connect fine-tuned, expert AI models, allowing the execution of complex tasks through plain language interfaces. ToxPipe utilizes Azure Cloud to enhance the speed and efficiency of toxicological predictions, leveraging its pay-as-you-go model for cost-effective utilization of resources and direct access to crucial OpenAI models. The project also integrates Posit Connect for the deployment of Shiny web applications, Jupyter notebooks, and APIs, allowing seamless interaction with ToxPipe through a primary Shiny application and facilitating the development of additional applications using the ToxPipe API. The project leverages existing resources on BioWulf and NIEHS HPC clusters for heavy compute loads, exploring hybrid deployment models and reducing cloud costs. The utilization of these resources also allows easy integration of scientific tools and data already loaded on BioWulf. The implementation of ToxPipe is poised to significantly accelerate the research process in toxicology, serving as a blueprint for harnessing cloud computing in scientific discovery. The project's success will be evaluated against a set of clearly defined metrics and performance indicators, including accuracy, speed, cost-effectiveness, and the quality and scope of ToxPipe generated products, aligning with NIH's strategic goals in data science and research. In conclusion, ToxPipe represents a groundbreaking initiative in computational toxicology, combining expert entrained AI, cloud computing, and hybrid deployments to advance toxicological workflows and scientific knowledge discovery. The well-defined milestones and experienced team are set to ensure the project's success and contribute to the evolution of toxicological research methodologies.

Keywords: retrieval-augmented generation, generative ai, large language models, toxicology, autonomous agents

1. Introduction

- Background on toxicological information retrieval.
- The significance of AI in enhancing toxicological data interpretation.
- Objective of the ToxPipe project.

Toxicological research is an important field that plays a critical role in protecting public health and the environment. However, the sheer volume of toxicological data available can make it difficult for researchers to quickly and accurately retrieve and interpret the information they need. ToxPipe is designed to address this challenge by providing an AI-assisted platform that can help researchers

quickly and accurately retrieve and interpret toxicological information.

2. Materials and methods

ToxPipe uses a combination of natural language processing, machine learning, and data visualization techniques to help researchers retrieve and interpret toxicological information. The platform is designed to be user-friendly and intuitive, with a simple and intuitive interface that allows researchers to quickly and easily search for and retrieve the information they need.

2.1. Data Collection

Description of the data sources and collection methods.

2.2. AI Model Development

- Overview of the AI models used.
- Training and validation of the models.

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2.3. Platform Design

Description of the ToxPipe platform’s user interface, features, and functionalities.

3. Results

ToxPipe has been tested on a variety of toxicological datasets, and has been shown to be highly effective at retrieving and interpreting toxicological information. The platform has also been well-received by toxicologists and researchers, who have praised its ease of use and effectiveness.

3.1. Model Performance

- Metrics showcasing the performance of the AI models.
- Comparative analysis with existing systems, if any.

3.2. Platform Usability

Feedback and results from user testing of the ToxPipe platform.

4. Discussion

- Interpretation of the results.
- Implications of the findings.
- Potential applications and impact of the ToxPipe platform in the field of toxicology.

ToxPipe represents a significant advance in the field of toxicological research, providing researchers with a powerful tool for quickly and accurately retrieving and interpreting toxicological information. The platform has the potential to significantly improve the efficiency and effectiveness of toxicological research, and could have important implications for public health and the environment.

5. Conclusions

Summary of the main findings and future directions for the ToxPipe project.

6. Acknowledgements

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7. Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used ChatGPT (GPT-4) in order to generate an initial abstract draft from the contents of a funding proposal for this project.

Additionally, ChatGPT (GPT-4) was used to generate an initial outline for this manuscript, which was then heavily edited by the author(s) and expanded upon.

After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

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